CS2008: Numerical Computing
Marks -------/10 Name & R#

Quiz 3: Least Square approximations

You are given the following toy dataset:

X	Y
3	10
4	15
5	27

Steps:
1. Fit the polynomial $p(x) = a_0 + a_1x + a_2x^2$ and find $\hat{y}$.
2. Calculate the error: $\Sigma(y - \hat{y})^2$.

Solution

As already mentioned, that for ***m*** degree polynomial,

The resulting linear system is of order $m + 1$. Specifically, the coefficients a_j, $j = 0, 1, \ldots, m$ can be found by solving the linear system

$$\mathbf{Az} = \mathbf{b},$$

where $\mathbf{z} = (a_0, a_1, \ldots, a_m)^T$,

$$\mathbf{A} = \begin{pmatrix} N+1 & \sum x_i & \sum x_i^2 & \cdots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \cdots & \sum x_i^{m+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum x_i^m & \sum x_i^{m+1} & \sum x_i^{m+2} & \cdots & \sum x_i^{2m} \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \\ \vdots \\ \sum x_i^m y_i \end{pmatrix}.$$

Matrix Form:

$$\begin{bmatrix} 3 & 12 & 50 \\ 12 & 50 & 216 \\ 50 & 216 & 962 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 52 \\ 225 \\ 1005 \end{bmatrix}$$

Using a calculator, we find:
$a_0 = 37$, $a_1 = -19.5$, $a_2 = 3.5$

Therefore, the fitted polynomial is:
$p(x) = 37 - 19.5x + 3.5x^2$

We can now calculate the fitted values ($\hat{y}$):

X	Y	$p(x) = \hat{y}$
3	10	10
4	15	15
5	27	27

Since $\Sigma (y - \hat{y})^2 = 0$, the polynomial fits the dataset perfectly.
This indicates that the given polynomial p(x) exactly matches the dataset.